

IoT-Based Wearable System for Real-Time Stress Detection & Adaptive Mental Health Support



Introduction & Problem Statement

Rising Stress Crisis

Stress and anxiety are leading causes of mental health disorders globally, often going undetected until severe.

Lack of Real-Time Monitoring

Traditional diagnosis relies on self-reporting — slow, subjective, and often inaccurate.

The Need

A continuous, objective, wearable solution that detects stress as it happens and responds adaptively.



**1B+People
Affected
Mental
health
disorders
worldwide**

**76 % Go
Untreated
Inlow- and
middle-
income
countries**

System Overview

A multi-sensor wearable platform powered by **ESP32** Microcontroller for real-time physiological data acquisition and processing.



Heart Rate Sensor

Monitors BPM continuously to detect cardiovascular stress indicators.



GSR Sensor

Measures galvanic skin response — skin conductance rises under stress.



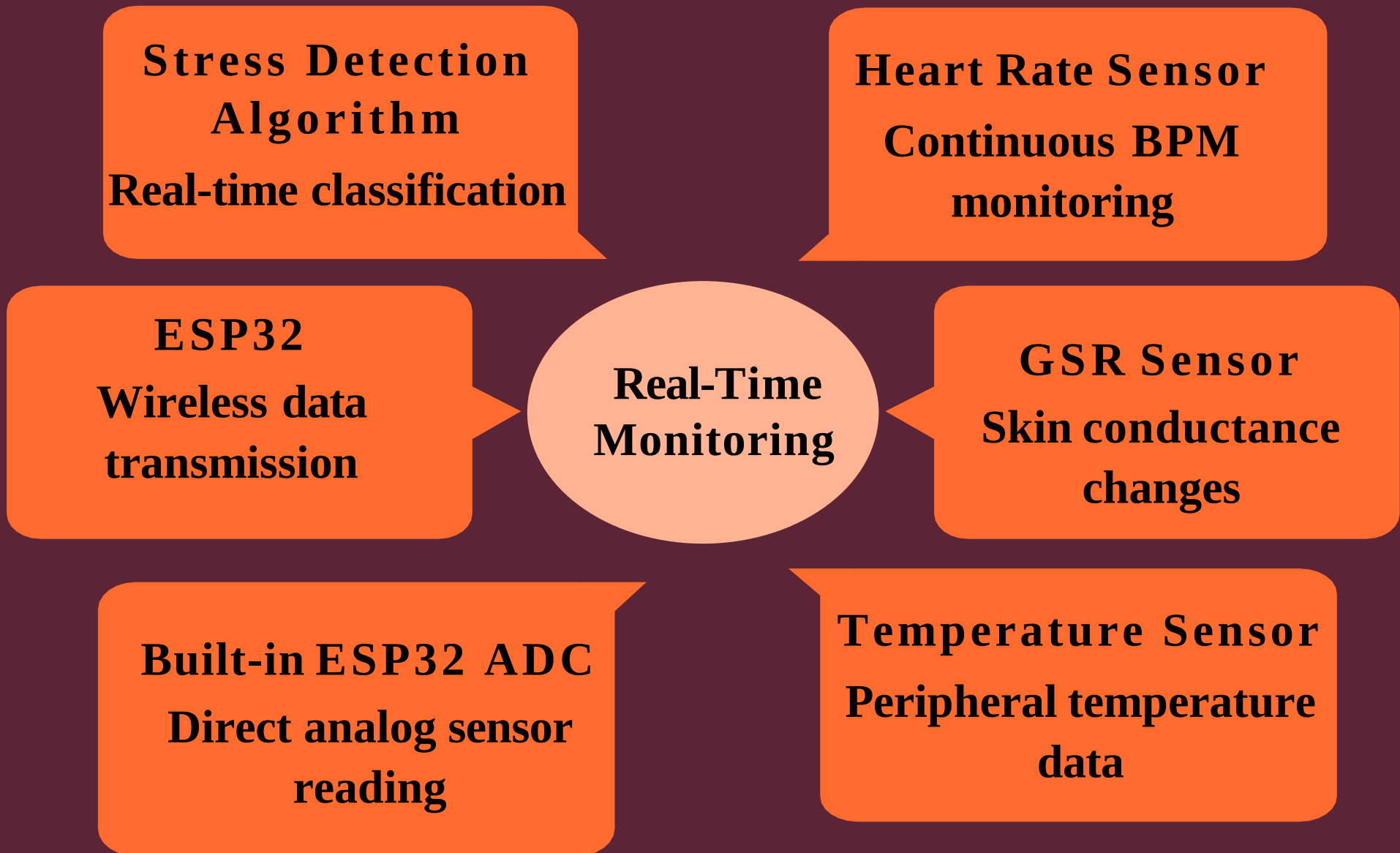
ESP32 Microcontroller

Central processor with built-in ADC for directly reading analog sensor values. Features Wi-Fi and Bluetooth connectivity.

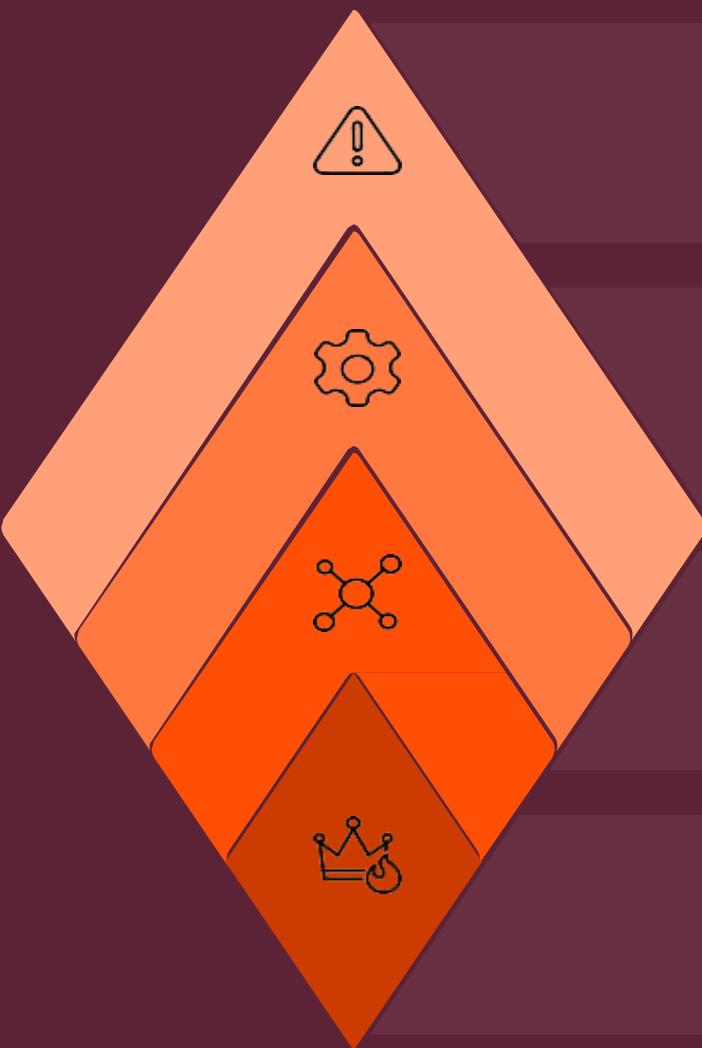


Temperature Sensor

Tracks body temperature fluctuations linked to emotional arousal.



Working Principle



01 Detect Stress?
Classify with algorithm

02 Preprocess Signals
Filter and normalize

03 Collect Sensor Data
Sample HR, GSR, temp

04 Start
Initialize sensors

01

Sensor Data Collection

Heart rate, GSR, and temperature sampled continuously from the wearable.

02

Signal Processing

ESP32 processes sensor data directly using its built-in ADC and sends data wirelessly for monitoring.

03

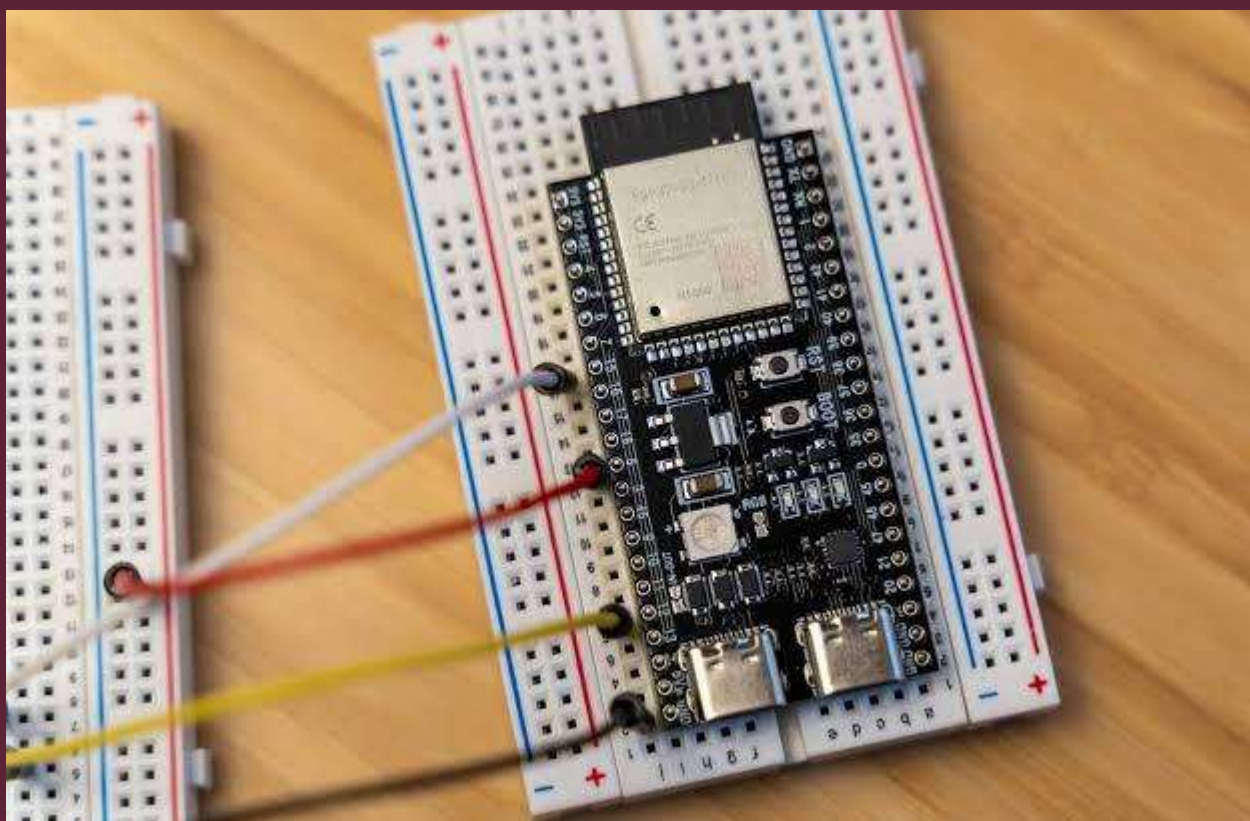
Stress Detection

Algorithm classifies stress level using threshold-based or ML logic.

04

Adaptive Response

System triggers calming interventions or emergency alerts based on severity.



Post-Detection Response System



Immediate Alert

Notification, LED, or buzzer indicates high stress detected



Calming Response

Plays soothing music and guides breathing exercises (inhale-exhale pattern)



Emergency Alert

Sends message or notification to caregiver if stress is critical



Data Logging

Stores stress data for future analysis and personalization

Features & Applications



Real-Time Monitoring

Continuous physiological tracking with instant stress-level feedback.



Calming System

Automatically plays soothing music, guides breathing exercises, or displays calming visuals.



Emergency Alerts

Notifies caregivers or emergency contacts when critical stress thresholds are exceeded.



ESP32 Advantages section: Wi-Fi connectivity, Bluetooth support, low power consumption, compact and cost-effective hardware



Applications

Healthcare, corporate wellness, military, students, and elderly care environments.

Conclusion & Future Scope

Project Summary

This system successfully integrates **IoT sensors, edge computing, and adaptive response** into a wearable platform for real-time, objective stress detection — bridging the gap between technology and mental health care.

✓ **Non-invasive
continuous**

✓ **Adaptive &
responsive**

✓ **Low-cost &
scalable**

Future Improvements

- Integrate **machine learning** models for personalized stress profiling
- Add **EEG and SpO₂ sensors** for richer physiological data
- Cloud-based **data analytics dashboard** for clinicians
- Miniaturize hardware into a **dedicated wearable chip**

THANK YOU